



A.M.($\leq$)	Any	Moon P.	Dark
Image Quality		Any seeing os	
Cloudy Cover		Photometric	

Title of Proposal: Rotation light curves of the Jupiter's Trojan asteroid Diomedes, Nestor and Antilochus

Telescope 0.60m Boller & Chivens

Abstract:

Jupiter's Trojan asteroids preserve their primordial physical characteristics since they are located at the Lagrangian points L4 and L5 of this planet. Therefore, these asteroids have undergone rare collisional events, which allows us to restrict the formation and evolution of the Solar System models. Currently, there are more than 12 thousand known Trojans, but only approximately 14% have their sizes estimated. Thus, it's of great importance to gather precise physical characteristics, both in size and shape, starting with the largest one and progressively moving to the smallest. The asteroids Diomedes, Nestor, and Antilochus are among our targets, which have diameters of ~118 km, ~112 km, and ~109 km, respectively, obtained by the NEOWISE satellites. Observing the rotation light curves of these bodies in the 2026A semester will contribute to determining their rotational phases at the predicted occultation instants for 2026.

Time requested: 180 hours.

Minimum time accepted: 120 hours.

Observing Mode: Classical Remote (CR)

Program Type: Regular Proposal (RP)

Principal Investigator (PI): Hélio Honório Dutra

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Is there a thesis or dissertation that will benefit from the data?: YES

Student	E-mail	Degree	Defense
Hélio Honório Dutra	hdutra@ov.ufrj.br	PhD student	02/2028

Optimal dates:

03/2026 to 05/2026

Impossible dates:

01/03/2026 to 12/03/2026

Instruments:

Instrument	Camera
Imager (CAM 1; 2 or 4)	Ikon 2048 x 2048

Instrument	Filter	Grating	Slit
Imager (CAM 1; 2 or 4)			
	UBVRI		

Scientific Justification

The Trojans are the part of a group of asteroids located in Jupiter's Lagrange regions, L4 and L5, where the gravitational and centrifugal forces of this planet and the Sun combine, creating a gravitational equilibrium [1] maintaining the relative positions of the Trojans to these two bodies. The Lagrangian points L4 and L5 are 60 degrees away from a planet, with L5 before and L4 after the planet in the direction of its orbital motion. Trojan objects tend to remain in these locations indefinitely with rare collisional events, maintaining their structures almost intact since their formation [2]. It is estimated that their population is as numerous as that of asteroids in the Main Belt, but only slightly more than 12 thousand objects are known. Only about 14% of them have measured sizes, usually in the infrared by space probes and telescopes [3]. Thus, there is a great need for surveying precise physical characteristics, both in size and shape, starting with the largest and progressing to the smallest.

Observing the rotational light curve of these bodies in the first semester of 2026 will contribute to determining the geometric albedo, using the definition adopted in the work of [4], at the times of the stellar occultations predicted for the next semester. Scaling the initially dimensionless 3D models into the stellar occultation chords, as was done in the work we recently published [5] of the Trojans Diomedes and Nestor available in DAMIT (Database of Asteroid Models from Inversion Techniques), allows us to calculate the equivalent radius in km of these bodies and, consequently, derive the values of their geometric albedos with great precision from the determination of the body's rotational phase at the instants of occultation event and the more precise reduced apparent magnitude obtained from these light curves.

There is still no 3D model of Antiochus in the database, but the rotation light curve of this body that will be obtained, together with the light curves obtained in semester 2025A at the OPD (still under analysis) and with those available in the ALCDEF (Asteroid Lightcurve Photometry Database), will be important for the construction of the 3D model using the light curve inversion technique [6], since these data are well spaced along the Phase Angle Bisector Longitude (PABLon), i.e., they were seen at different aspect angles and solar phase angles along its orbit (see figure 1a in the appendix). Despite few rotation observations of Diomedes and Nestor available in the ALCDEF database (see figures 1b and 1c in the appendix), the observation will contribute to the future construction of their more detailed 3D models.

Currently, there are predictions of stellar occultations for the first half of 2026 for these three targets of the proposal. Stellar occultations for Diomedes: (14/04/26: <https://abrir.link/EnHfk>); (19/04/26: <https://abrir.link/NlZOk>) and (25/04/26: <https://abrir.link/VNaWe>). For Nestor: (05/03/26: <https://abrir.link/lmoCG>); (21/03/26: <https://abrir.link/HOMOE>) and (15/04/26: <https://abrir.link/oKIEE>). For Antiochus: (21/03/26: <https://abrir.link/VRAvh>); (31/03/26: <https://abrir.link/KrEXB>); (16/04/26: <https://abrir.link/mzmUi>) and (13/06/26: <https://abrir.link/IHLnP>).

These targets will have more visibility during the night in the months of March, April, and May, periods in which the stellar occultations listed above will occur.

This project is a part of a PI's PhD thesis whose objective is to accurately determine the size and shape, as well as other physical characteristics, of Trojan asteroids by combining rotation light curve analysis, light curve inversion technique and stellar occultation events. It is one of the projects that are part of the Rio Group, led by Dr. Roberto Martins.

References

- [1] Gunter Faure, Teresa Mensing. Introduction to Planetary Science. Springer, 2007.
- [2] Bottke, W. Asteroids III. Space Science Series. University of Arizona Press, 2002.
- [3] Grav. T.; et al. WISE/NEOWISE Observations of the jovian trojans: preliminary results. The Astrophysical Journal, v. 742, p. 40, 2011.
- [4] Sicardy B et al. 2011 A Pluto-like radius and a high albedo for the dwarf planet Eris from an occultation. Nature 478, 493–496.
- [5] Dutra H et al. 2025 Physical characteristics of Jupiter's Trojan (1437) Diomedes from a tri-chord stellar occultation in 2020 and dimensionless three-dimensional model. Phil. Trans. R. Soc. A 383: 20240187.
- [6] Kaasalainen M., Torppa J., 2001, Icar, 153, 24.

Experimental Design

For the first step, we need at least a total of 10 nights divided into 2 blocks of 5 nights each (which can be continuous or distributed in intervals of 1 or 2 nights), with a 20-day interval between these blocks, starting on March 13th. The other 5 nights can be in the period from May 11th to May 21st, after these two blocks, and can be continuous or distributed in intervals of 2 or 3 nights. We will use the Clear and I filters. We can use other filters for backup projects. We would like the IkonL camera in order to have a wider field of view for photometric and astrometric comparison stars.

During observation, we will use 60 to 120 seconds of exposure time, possibly reaching 200 seconds depending on weather conditions and the air mass. Bias and Flat Field calibration images will be acquired at the beginning or end of each night. The treatment method will be aperture differential photometry on the target that's moving in the field of view, using the stars in that field as calibrators. The result will be a rotation light curve for each target, determining its rotational period. We will use all available time to alternate targets and thus ensure that all are observed. From these light curves, it will be possible to obtain the asteroid's rotational phase at the instant of the occultation, as well as its reduced apparent magnitude. Having the value of its diameter with good precision by scaling the dimensionless 3D model on the occultation chords, we will be able to calculate its geometric albedo with an accuracy of 0.001. The derived rotation light curves will also be implemented to construct more detailed 3D models.

Technical Description

The characteristics of the requested telescope (0.6 m aperture B&C) are sufficient for detecting the brightness variation of the targets, as they have apparent magnitudes between 15 and 16. This allows for determining the amplitudes of the rotation light curves due to the rather irregular shape of these objects. However, due to the known orientation of the pole and the synthetic light curve for the Trojans Diomedes and Nestor (available at DAMIT) for the selected observation period, the relative amplitude of the predicted flux ratio is approximately 0.5. For this reason, the nights close to the new moon were requested in order to mitigate the sky background brightness. Thus, an SNR > 30 is sufficient to detect the variation in brightness due to body rotation, considering their apparent magnitude. An exposure time from 60 to 120 seconds is quite enough to obtain at least this SNR.

The best period for observing the targets is between March 1st and May 31st, when they will be visible for longer in the night sky in the southern hemisphere. The rotation period of the asteroid Diomedes is $P \sim 24.49$ hours. This means that almost the same face of Diomedes will be observed for several consecutive nights, making it difficult to calculate its complete rotation. To circumvent this, I requested a 20-day interval between observation blocks of 5 nights each, so that the 0.49-hour night lag phase is accumulated, allowing the other face to be observed after this idle interval: a difference of approximately 150° in relation to the first observation block. The targets will be visible for about 6 to 10 hours during the requested period (the longest possible time during the next semester). Thus, in a 5-night block, it will be possible to cover the period of another target, which could be Nestor ($P \sim 15.98$ h) or Antiochus ($P \sim 15.76$ h), that is, one of the aforementioned night blocks will be sufficient to cover these last two alternately.

Previous missions of this proposal to this telescope

Data from other Trojan asteroids Agenor, Alcahous, and Priamus continue to be collected at the OPD in this semester following the approval of the previous application for semester 2025B, and are being analyzed and processed to obtain the results following, in part, the methodology described above.

We also collected rotational light curves of the same targets of this proposal and those data are under analysis using the methodology described above. It's worth noting that we already have 7 occultation events by Diomedes, 3 by Nestor, and 8 by Antilochus being reduced in the moment.

Previous missions of other proposals to this telescope

N/A

Previous results in field by the Principal Investigator (PI)

The PI and co-authors scaled the Trojan Diomedes model using through from a tri-chord stellar occultation in 2020 and refine the existed pole coordinate pole ($\lambda = 153.73^\circ \pm 2.5^\circ$, $\beta = 12.69^\circ \pm 2.6^\circ$) in the ecliptic coordinate system and derived the new rotation period ($P = 24.4984 \pm 0.0002$ h). From that, they derived the geometric albedo ($p_V = 0.030 \pm 0.004$) and volume-equivalent radius ($R_{eq} = 59.4 \pm 0.3$ km). The precise astrometric position was also determined. These results were published in the article Dutra H et al. 2025 (<https://doi.org/10.1098/rsta.2024.0187>).

PI and Co-Is publications relevant to this proposal

Dutra H et al. 2025 Physical characteristics of Jupiter's Trojan (1437) Diomedes from a tri-chord stellar occultation in 2020 and dimensionless three-dimensional model. Phil. Trans. R. Soc. A 383: 20240187. <https://doi.org/10.1098/rsta.2024.0187>

Ferreira, F. S., Camargo, J. I., Morgado, B. E., Peixoto, V. F., Desmars, J., Braga-Ribas, F., ... & Wagner, K. (2025). Size and Shape of Jupiter Trojan (2207) Antenor from Stellar Occultations. The Astronomical Journal, 170(3), 160.

Pereira, C. L., Braga-Ribas, F., Sicardy, B., Leiva, R., Assafin, M., Morgado, B. E., ... & Vanzi, L. (2025). The rings of (2060) Chiron: Evidence of an evolving system. The Astrophysical Journal Letters, 992(2), L19.

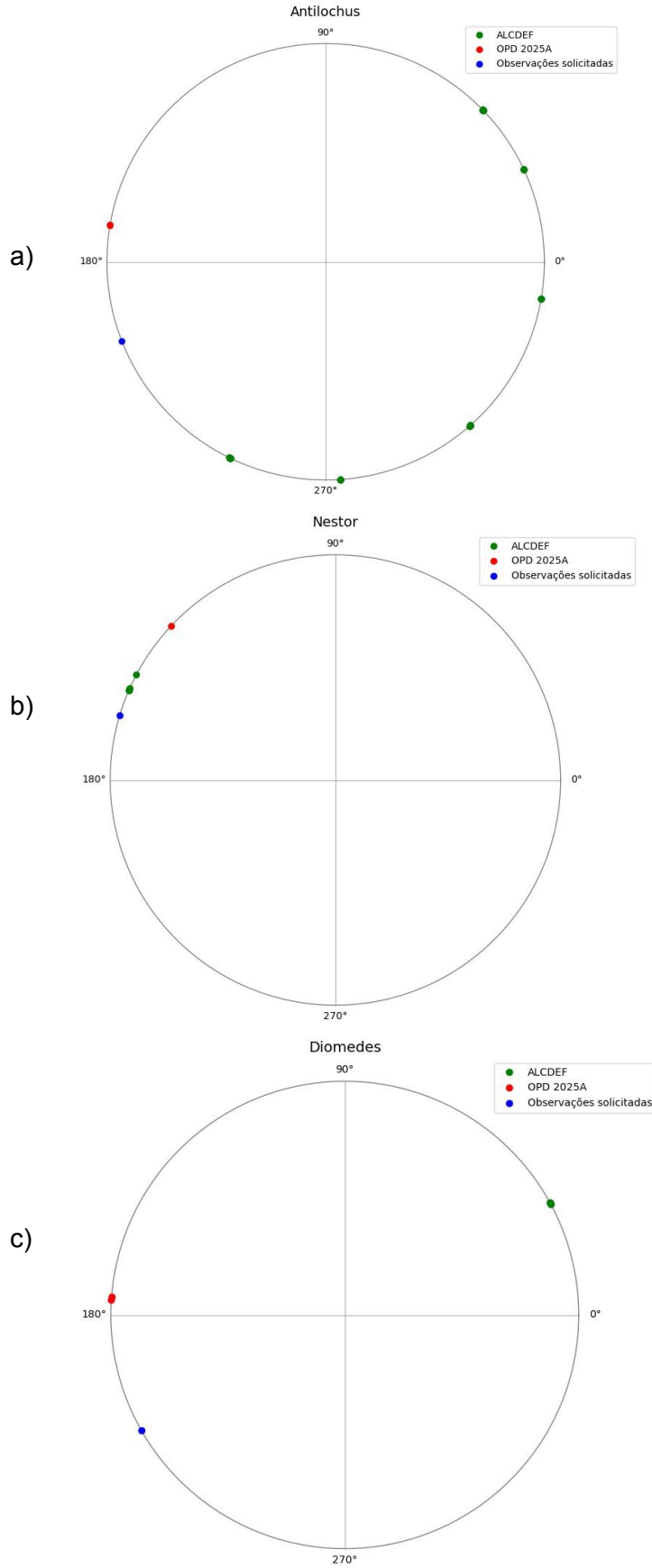


Figure 1: Distribution of the observed rotation light curves of a) Antilochus, b) Nestor and c) Diomedes, both available in the ALCDEF database (in green), obtained at the OPD (in red) and from observations requested for semester 2026A (in blue) along the phase angle bisector longitude (0° and 90° respectively to the right and above). The more sparse is the observation along the orbit, more precise the rotation curves inversion for the construction of the 3D model become.

Objects List.

Object	A.R.(h,m)	Dec.(deg)	Mag.	Band	Type	Comments
(1437) Diomedes	13:26:13.72	-33:47:54.4	15.605	V	Sci	Position on March 18th, 2026 at 03:00 UTC
(659) Nestor	10:54:10.42	+07:38:33.0	16.183	V	Sci	Position on March 18th, 2026 at 03:00 UTC
(1583) Antilochus	13:05:03.46	-22:15:26.7	15.905	V	Sci	Position on March 18th, 2026 at 05:00 UTC